Dr. AMBEDKAR INSTITUTE OF TECHNOLOGY

(An Autonomous Institute, Affiliated to Visvesvaraya Technological University, Belagavi, Accredited by NAAC, with A+ Grade)

Near Jnana Bharathi Campus, Bangalore – 560056

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Dissertation Report on

"A THREE LAYER BASED INTELLIGENT DATA PRIVACY PROTECTION SCHEME IN CLOUD STORAGE"

Submitted in partial fulfillment of the requirement for the award of the Degree of

Master of Technology
in
Computer Science and Engineering

Submitted by

RANJITHA R

1DA23SCS12

For the academic year 2024-2025

Under the Guidance of

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2024-2025

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CERTIFICATE

This is to certify that the Project entitled **"A Three Layer based Intelligent Data Privacy Protection Scheme in Cloud Storage"** carried out by **RANJITHA R** bearing USN **1DA23SCS12**, a bonafide student of **Dr. Ambedkar Institute of Technology** in partial fulfillment for the award of the degree **Master of Technology in Computer Science & Engineering** under **Visvesvaraya Technological University, Belgaum** during academic year 2024–2025. It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated in the report deposited in the departmental library. The Project report has been approved as it satisfies the academic requirements in respect of Project work prescribed for the said degree.

Signature of Internal Guide
Dr. Siddaraju
Vice-Principal, Professor
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Signature of Co-Ordinator
Dr. Shylaja K R,
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11/9/25

Signature of HOD
Dr. Nandini N,
Professor and Head
Dept of CSE, Dr. AIT
Bangalore.

Signature of Principal
Dr. M N Thippeswamy,
Principal
Dr. AIT,
Bangalore.

Viva- Voice Examination
Name of the Examiners

Signature with Date

1) _____

2) _____

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DECLARATION

I hereby declare that the matter embodied in this report is the result of project work carried out by me at the Department of Computer Science and Engineering in Dr. Ambedkar Institute of Technology, Bengaluru under the supervision of Dr. Siddaraju, Vice-Principal, Professor, Department of Computer Science Engineering. The dissertation report or part of has not been submitted for the award of any degree, diploma, fellowship etc., of any other University or Institute.

Bengaluru – 560056 **RANJITHA R**

Date: **1DA23SCS12**

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ABSTRACT

The rapid adoption of cloud storage services necessitates robust mechanisms for ensuring data privacy and security. This project presents a comprehensive three-layer intelligent data privacy protection scheme tailored for cloud storage, implemented using Java for backend processes, and JSP, HTML, CSS, and JavaScript for the frontend interface. The database management is handled using MySQL, ensuring efficient data storage and retrieval.

Our system leverages the AWS Cloud service provider for cloud storage solutions. It introduces a controlled user registration process where new users can only access the system upon administrative approval, enhancing security by preventing unauthorized access. Once registered, users can upload files to the cloud, where the system employs a multi-faceted approach to data protection.

Uploaded files are partitioned into three distinct blocks, each encrypted using the Caesar Cipher algorithm. This classical encryption method, despite its simplicity, provides a foundational layer of security. Additionally, our system incorporates a deduplication feature that identifies and manages redundant data uploads, thereby optimizing storage efficiency.

To ensure secure access and retrieval, a unique Message Authentication Code (MAC) is generated for each of the three blocks of data. This MAC Code is essential for the decryption process, ensuring that only authorized users can reconstruct and access the original files. The combination of data splitting, encryption, and MAC-based authentication forms a robust framework for maintaining data confidentiality and integrity in cloud environments.

This project demonstrates a practical approach to enhancing data privacy in cloud storage systems, offering a scalable and secure solution for both individual users and organizations.